



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SHRI SHANMUGA ENTERPRISES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

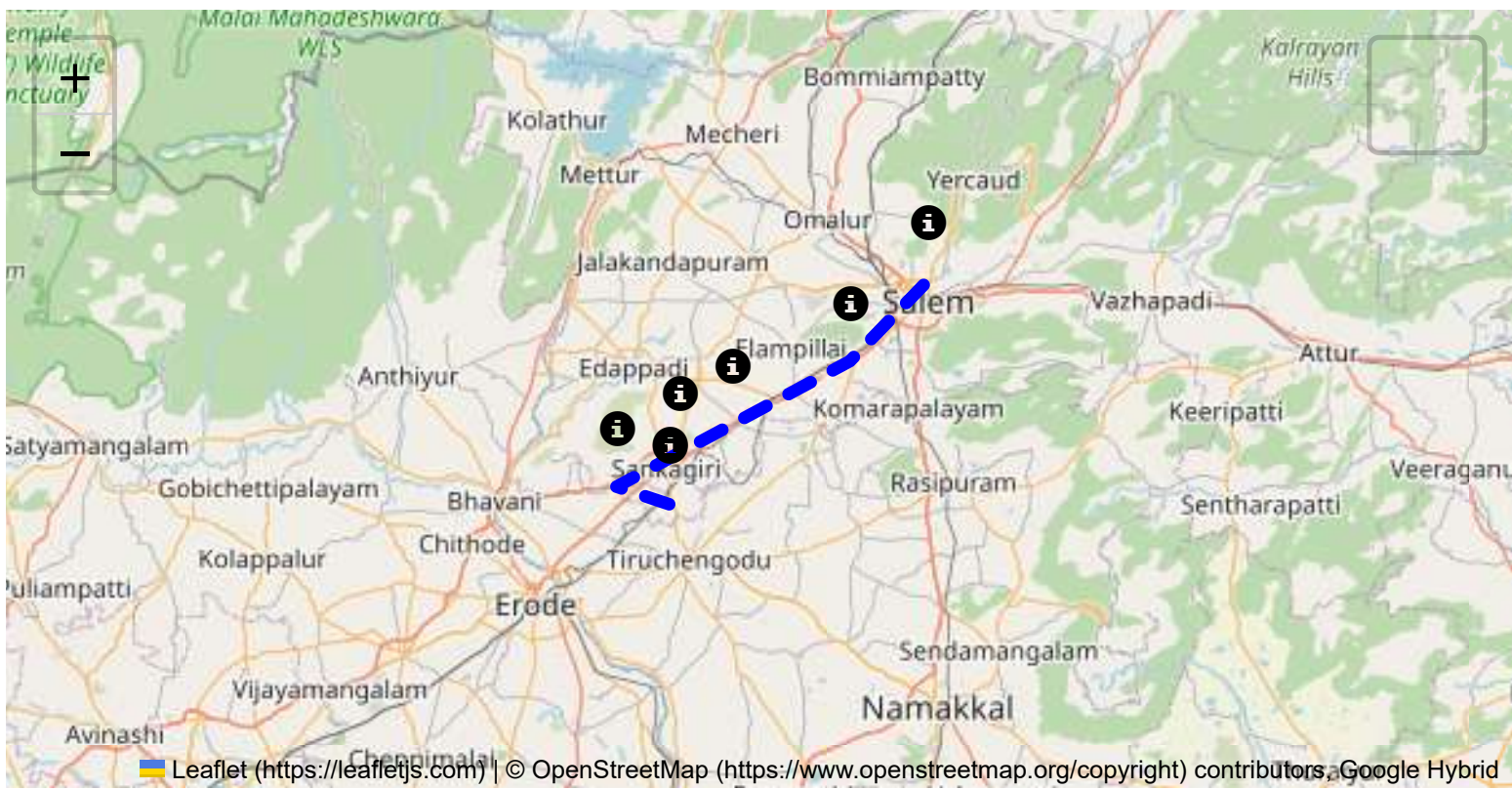
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

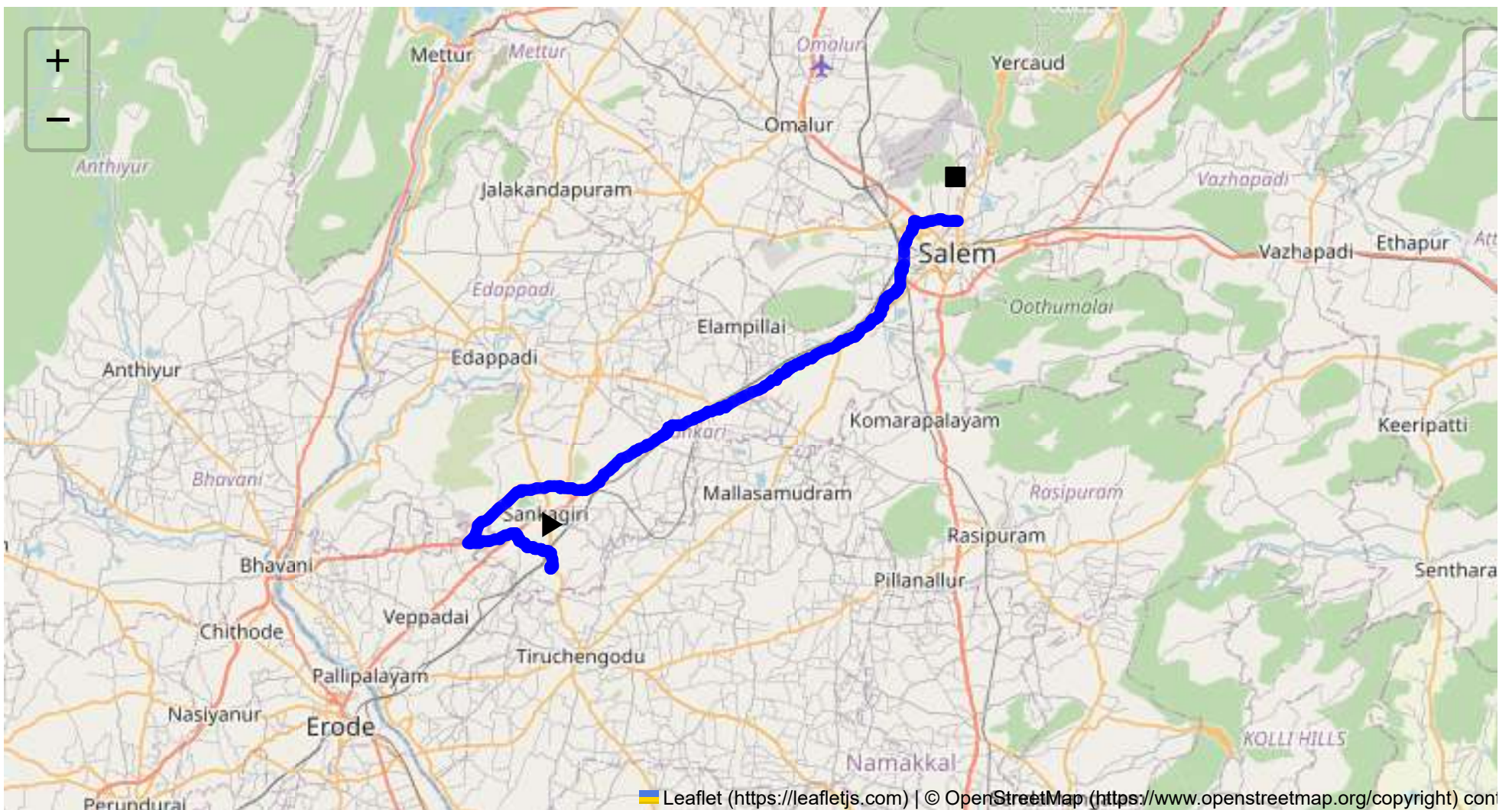
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 57.20 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.675845, 78.15551)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri, Tamil Nadu to Sarada College Road in Salem, Tamil Nadu, covers approximately 57.20 kilometers. It passes through key waypoints, including Puthur, Bhavani Main Road Padaiveedu, Salem - Kochi Highway, Vaikundam, and the Salem Bypass. It is vital to travel through various regional roads to access Salem, a significant city in Tamil Nadu.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. Around this route, the weather is usually hot and humid. The southwest monsoon season, typically from June to September, can bring heavy rainfall which might lead to waterlogged roads and poor visibility, posing hazards. Additionally, the northeast monsoon between October and December can also result in similar conditions.

3. Traffic Patterns and Congestion-Prone Areas

Traffic is usually moderate to heavy, especially during weekdays. Peak hours often occur in the morning between 7:30 AM to 10:00 AM and in the evening between 5:00 PM to 8:00 PM. Congestion is likely near major town junctions and marketplaces, notably on the Salem Bypass and as you approach Salem city.

4. Assessment of Road Quality and Infrastructure

- **CVQF+23W to CVQF+VG:** Rural roads may have uneven surfaces, especially after monsoon rains.
- **CVQF+VG to FR48+7XQ:** Generally decent road conditions but with occasional potholes.
- **FVVM+MCQ to GWFR+CWP:** Is part of the Salem - Kochi Highway, offering relatively better road quality and multi-lane sections.
- **GWFR+CWP to Salem:** The Salem Bypass is well-maintained but can be busy.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, an alternative option could be to reroute via the NH 79 and then connect through NH 44 towards Salem. This could help mitigate congestion or road closures during unforeseen situations.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu strictly regulates the transportation of hazardous materials. Vehicles carrying such materials are often restricted to certain times and routes to minimize public exposure due to safety concerns. Drivers must comply with local laws and carry relevant documentation.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been occasional incidents involving heavy vehicles, particularly related to brake failures on steep gradients and accidents due to poor weather conditions. However, no specific pattern of hazardous materials incidents has been distinctly noted.

8. Environmental Considerations and Sensitive Areas

This route crosses several agricultural zones that can be environmentally sensitive, demanding caution to prevent any spillage. It's crucial to preserve local habitat and prevent pollution of any nearby water bodies or farmlands.

9. Analysis of Communication Coverage

Most segments of the route have good mobile network coverage. However, rural segments, especially around Vaikundam, might experience intermittent coverage issues, which could affect communication.

10. Estimated Emergency Response Times for Different Route Segments

- CVQF+23W to CVQF+VG: 20-30 minutes response time due to rural access.
- FR48+7XQ to FVVM+MCQ: 15-25 minutes with better road access.
- GWFR+CWP to Salem: 10-15 minutes, owing to proximity to urban emergency services in Salem.

12. Overall Summary of Risk Assessment

The route carries moderate risks mainly due to weather and road conditions. Regular checks on vehicle health to manage the rugged terrain, especially during monsoons, should be prioritized. Compliance with regulated hours for transporting hazardous materials remains vital. Alternative routes should be pre-planned to manage delays efficiently. Moreover, ensuring consistent communication availability en route is crucial to timely emergency management.

This analysis highlights critical aspects of the journey, offering guidance for safe travel and efficient incident response for truck drivers navigating this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.6762303, 78.1575524	10 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44898, 77.87410	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46048, 77.85036	30 KM/Hr
9	Turn	Medium	11.46303, 77.84945	30 KM/Hr
10	Turn	Medium	11.46318, 77.84913	30 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.45631, 77.81668	15 KM/Hr
13	Turn	High	11.49409, 77.88004	15 KM/Hr
14	Turn	Medium	11.49419, 77.88015	30 KM/Hr
15	Turn	Medium	11.49365, 77.88596	30 KM/Hr
16	Turn	High	11.49235, 77.89561	15 KM/Hr
17	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
18	Turn	Medium	11.64183, 78.11810	30 KM/Hr
19	Turn	High	11.67074, 78.12668	15 KM/Hr
20	Turn	High	11.67081, 78.12661	15 KM/Hr
21	Turn	High	11.67582, 78.12786	15 KM/Hr
22	Turn	High	11.67623, 78.15755	15 KM/Hr
23	Blind Spot	Blind Spot	11.67615, 78.15756	10 KM/Hr
24	Turn	High	11.67603, 78.15558	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
15	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
17	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
19	hospital	Venkatesh Hospital and Urology Centre	11.6759517, 78.1324029	30 km/h	Medium
20	hospital	R.K. Hospital, Salem	11.6746905, 78.1339717	30 km/h	Medium
21	hospital	Kurinji Hospital, Salem	11.6771806, 78.1351224	30 km/h	Medium
22	clinic	Ayush Care Siddha Centre	11.6731195, 78.1351967	30 km/h	Medium
23	clinic	Bismillah Piles Clinic	11.6755285, 78.1343325	30 km/h	Medium
24	hospital	Bhavani Hospital	11.6745697, 78.1362683	30 km/h	Medium
25	clinic	Arunagiri Ent Nursing Home	11.6762823, 78.1369593	30 km/h	Medium
26	clinic	Srimaya Child Health Centre	11.6750852, 78.1370646	30 km/h	Medium
27	hospital	Gl Hospital	11.6742597, 78.1379092	30 km/h	Medium
28	clinic	Ayush Multispeciality Homoeopathic Clinic	11.6770982, 78.1395947	30 km/h	Medium
29	hospital	Balaji Skin Hospital	11.676166, 78.1379488	30 km/h	Medium
30	hospital	Dr. Sasi Eye Care	11.6743698, 78.1379234	30 km/h	Medium
31	police	Alagapuram Police station	11.6769953, 78.1422615	30 km/h	Medium
32	hospital	AP medical center	11.6774074, 78.1425988	30 km/h	Medium
33	hospital	M G Diabetes Speciality and Research Centre	11.6766146, 78.1422986	30 km/h	Medium
34	hospital	Nathan Super Speciality Hospital	11.6743874, 78.1417222	30 km/h	Medium
35	hospital	Devi Hospital	11.6786391, 78.1415731	30 km/h	Medium
36	hospital	Fairlands Eye Cure	11.677959, 78.1415897	30 km/h	Medium

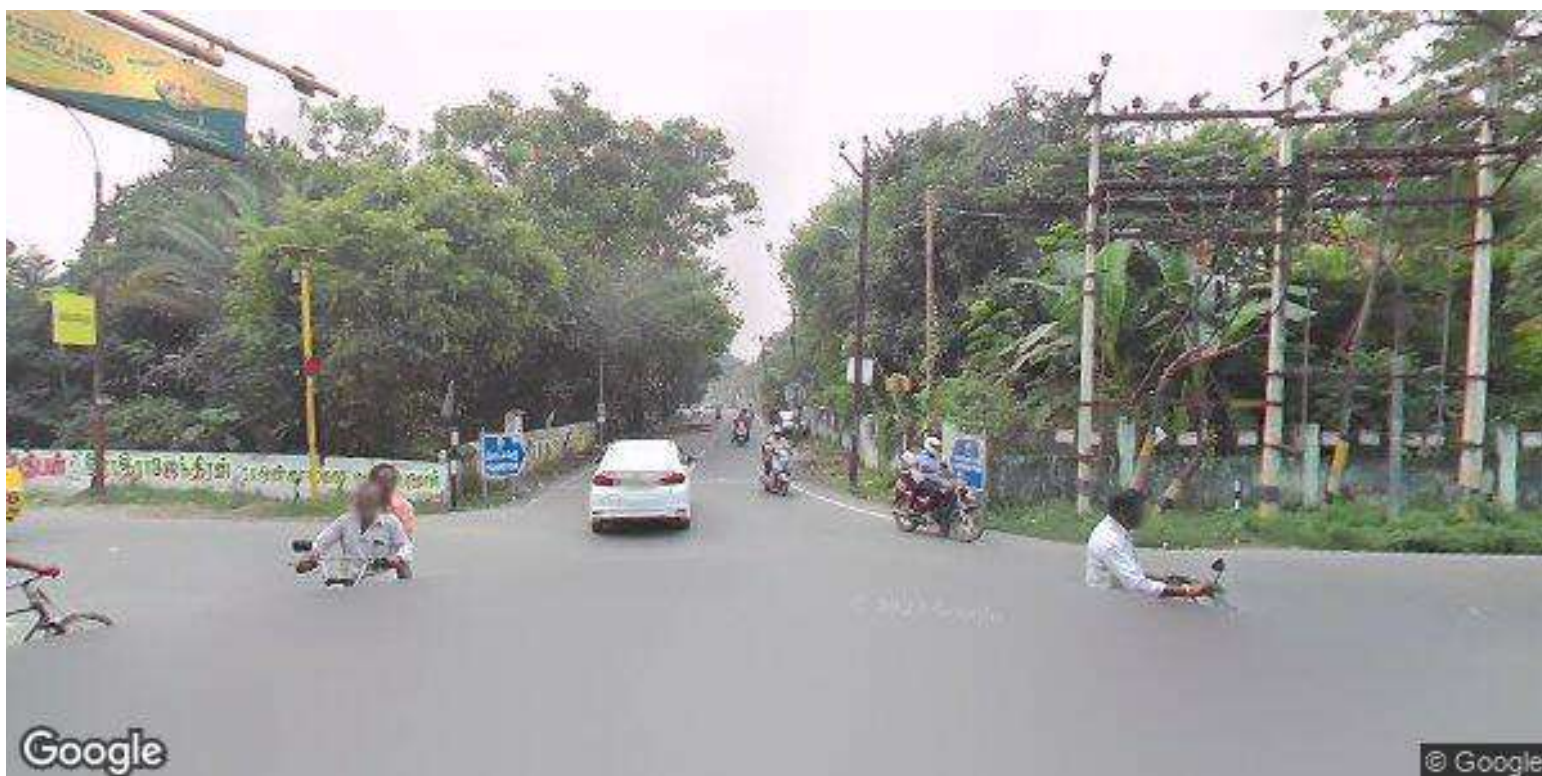
	type	name	coordinates	speed_limit	risk_level
38	hospital	Dr. Soodamani Hospital	11.6757031, 78.14473	30 km/h	Medium
39	hospital	Kavin Eniyan Hospital	11.677643, 78.144129	30 km/h	Medium
40	hospital	Sri Vaari Hospital	11.6763726, 78.1467277	30 km/h	Medium
41	hospital	Krishna Speciality Hospital	11.676751, 78.145561	30 km/h	Medium
44	hospital	Gopi Memorial Hospital	11.673855, 78.1516078	30 km/h	Medium
45	clinic	Saraswathi Nursing Home	11.6746327, 78.1520981	30 km/h	Medium
46	hospital	J S Hospital, Dharmapuri	11.6739064, 78.1521557	30 km/h	Medium
47	hospital	Salem Gopi Hospitals	11.6745894, 78.1515482	30 km/h	Medium
48	hospital	Dr. P. Sundarrajan Neuro Hospital	11.6742791, 78.1499738	30 km/h	Medium
49	hospital	Mahadevas Ayurvedic Hospital	11.6752045, 78.1517396	30 km/h	Medium
50	clinic	Akshaya Fertility Centre	11.6739088, 78.1518657	30 km/h	Medium
52	hospital	Shanmuga Hospital and Salem Cancer Institute	11.6770178, 78.1531844	30 km/h	Medium
53	hospital	Tms Eye Hospital	11.6772791, 78.1536202	30 km/h	Medium
54	hospital	Sureka Hospital	11.676423, 78.1550511	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
18	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
37	school	Saradha Balamandir School	11.6771182, 78.1472461	30 km/h	Medium
42	school	Saradha College	11.6769525, 78.1481901	30 km/h	Medium
43	school	Sri Sarada Matriculation Higher Secondary School	11.6768969, 78.1488115	30 km/h	Medium
51	school	Holy flower matric hr.sec school	11.6748637, 78.1528099	30 km/h	Medium
55	school	CSI Hobart Edu Institute	11.673245, 78.1567934	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.6762303, 78.1575524



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49365, 77.88596



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49235, 77.89561



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49225, 77.89560



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67074, 78.12668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67081, 78.12661



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67582, 78.12786



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67623, 78.15755



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.67615, 78.15756



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67603, 78.15558

